Correspondence

Direct Position Determination Using Compressive Sensing Measurements Without Reconstruction

This article proposes a direct position determination (DPD) method for stationary emitters with compressed signal measurements. Employing Hadamard matrix properties, the method estimates the source location directly in the compressing measurement domain, without the step of signal reconstruction and time-differences-of-arrival parameter extraction. In the case of free-space propagation, an augmented DPD method is proposed to further utilize the position-related received signal strength information. In addition, the Cramér–Rao lower bounds for the emitter position of both DPD methods are derived for performance evaluation.

I. INTRODUCTION

The problem of locating a source through its emitted signals has been of considerable interest to both academic researchers and engineering practitioners [1], [2], [3], [4], [5], [6], [7], [8]. Most emitter localization methods consist of two steps. In the first step, the position-related parameters are extracted through the signals of several receivers. Then, in the second step, the estimated emitter's position should best fit the collection of the extracted parameters. In the first step, the time-differences-of-arrival (TDOA) is preferable as a generally accurate position estimation that can be achieved through several time-synchronized one-channel receivers [9], [10].

However, such merit of the TDOA-based localization method is obtained usually at the cost of large inter-receiver communication bandwidth. To address the issue, many studies have been conducted to make the TDOA-based localization methods feasible in case in which the communication resources are restricted, among which compressive sensing (CS) techniques are commonly used [11], [12], [13].

A standard CS problem consists of two steps: signal compressing and signal reconstruction. In the signal compressing step, a compressing matrix is designed to compress the original measurements into the measurements of much lower dimensions [14]. Then, in the signal reconstruction step, efforts are made to recover the original measurements [15], [16], [17]. However, the signal reconstruction step may induce a great computation cost and is not preferable for real-time applications. To handle the issue, Salari et al. [18] develop a CS-based framework for TDOA estimation

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that eliminates the signal construction step by employing a special property of the Hadamard matrix. The method in [18] is proved to be superior to the orthogonal matching pursuit reconstruction method [19] in terms of TDOA estimation accuracy and the focal underdetermined system solver reconstruction method [20] in terms of processing time.

Generally, CS-based localization approaches perform well at high signal-to-noise ratios (SNRs). However, their performance degrades rapidly when SNR becomes low, which is very severe in the case of high CS rates. In practice, a promising technique to handle rapid performance degradation in low SNRs is the direct position determination (DPD) paradigm [21], [22], [23], [24], [25], [26], [27], [28]. Different from the two-step methods, a DPD method locates the emitter directly from the received signals. A DPD method can utilize the underlying restriction that all observations should be consistent with the same emitter. In addition, the process of parameter covariance estimation is avoided, which may sometimes be difficult to handle.

To improve the performance of CS-based localization methods, especially at low SNRs, this article is devoted to an effective integration of CS and DPD techniques, which has not yet been reported to our best knowledge. Specifically, employing Hadamard matrix properties, we develop a one-step CS-based DPD method that eliminates the complex reconstruction step and the TDOA parameter extraction step, and estimates the source location directly in the compressive measurement domain. In addition, an augmented DPD method is proposed to further improve the localization accuracy in case of free-space propagation by utilizing the position-related information in received signal strength (RSS). The Cramér–Rao lower bounds (CRLBs) for the emitter position of both DPD methods are derived. Our simulation results show that the proposed CS-based DPD method performs comparably with the classical DPD method without CS when SNR is high. By choosing a reasonable compressing rate, the CS-based DPD method can also perform well when SNR is low. When the emitted signals are subject to free-space propagation, better performance can be achieved by the RSS-augmented CS-based DPD method.

II. SYSTEM MODEL

Consider a stationary source located at $\mathbf{u} = [x, y, z]^T$ and L stationary receivers located at $\mathbf{p}_l = [x_l, y_l, z_l]^T$, $l = 1, 2, \ldots, L$. The source starts emitting signals at time $t = 0$. The receivers are time synchronized. The time the signals take to propagate from the source to receiver l is

$$\tau_l = \|\mathbf{u} - \mathbf{p}_l\| \big/ c \tag{1}$$

where c is the speed of wave propagation. Given that the source signal is $s(t)$ at time t, then the signal received by receiver l at time t can be modeled as

$$z_l(t) = a_l(t)\, s(t - \tau_l) + w_l(t) \tag{2}$$

where $a_l(t)$ is the path attenuation factor and is usually approximately stationary over the signal observing period. Consequently, we can assume $a_l(t) \approx a_l$. The noise term $w_l(t)$ is zero mean and statistically independent circular complex Gaussian noise such that $\mathbb{E}\{w_l^2(t)\} = \sigma_l^2$. In practice, the continuous noise term $w_l(t)$ is mainly generated due to the thermal vibration of electrons of analog devices and analog-to-digital converters, and the dominant component is usually from the low-noise amplifiers. For the case of using the same type of receivers, which is common in practice, we assume $\sigma_1^2 = \cdots = \sigma_L^2 = \sigma_w^2$ in this article. Note that it is straightforward to extend to heterogeneous noise levels [29].

Let Δ be the sampling interval, and the sampled signal is

$$z_l[n] = a_l s(n\Delta + t_0 - \tau_l) + w_l[n] \tag{3}$$

where $w_l[n] = w_l(n\Delta + t_0)$ and t_0 is the initial moment of signal reception. Suppose that N sampling points are collected, and let the collected signal vector be

$$\mathbf{z}_l = [z_l[0], z_l[1], \ldots, z_l[N-1]]^T. \tag{4}$$

Also, let

$$\mathbf{s} = [s[0], s[1], \ldots, s[N-1]]^T \tag{5}$$

where $s[n] = s(n\Delta + t_0)$. Then, $\mathbf{z}_l$ can be modeled in terms of $\mathbf{s}$ through a fast Fourier transform (FFT) and inverse FFT treatment [26], [29]

$$\mathbf{z}_l = a_l \mathbf{F}^H \mathbf{D}_{\tau_l} \mathbf{F}\mathbf{s} + \mathbf{w}_l = a_l \mathbf{Q}_l \mathbf{s} + \mathbf{w}_l \tag{6}$$

where

$$\tilde{\mathbf{n}} = \left[-\frac{N}{2}, \ldots, -1, 0, \ldots, \frac{N}{2} - 1\right]^T \tag{7a}$$

$$\mathbf{F} = \frac{1}{\sqrt{N}} \exp\left(-j\frac{2\pi}{N}\tilde{\mathbf{n}}\tilde{\mathbf{n}}^T\right) \tag{7b}$$

$$\mathbf{D}_{\tau_l} = \operatorname{diag}\left\{\exp\left(-j\frac{2\pi}{N}\tilde{\mathbf{n}}\frac{\tau_l}{\Delta}\right)\right\}. \tag{7c}$$

In practice, the collected signal vector $\mathbf{z}_l$ is usually transferred to a central receiver for TDOA extraction leading to a demanding requirement on communication bandwidth. To reduce the burden on the communication link, CS techniques are used to compress the original sampled signals. In particular, we have the compressed signal received at the central receiver

$$\begin{aligned}\mathbf{y}_l &= \mathbf{\Phi}_l \mathbf{z}_l + \boldsymbol{\varepsilon}_l \\ &= a_l \mathbf{\Phi}_l \mathbf{Q}_l \mathbf{s} + \mathbf{\Phi}_l \mathbf{w}_l + \boldsymbol{\varepsilon}_l\end{aligned} \tag{8}$$

where $\mathbf{\Phi}_l$ represents the $M \times N$ compressing matrix with $M \ll N$, the term $\boldsymbol{\varepsilon}_l$ is the vector of noise generated in the communication process between receiver l and the central receiver, $\boldsymbol{\varepsilon}_l = [\varepsilon_l[0], \varepsilon_l[1], \ldots, \varepsilon_l[M-1]]^T$, where $\varepsilon_l[n]$ is the zero mean and statistically independent circular complex Gaussian noise and $\mathbb{E}\{\varepsilon_l^2[n]\} = \sigma_{\varepsilon l}^2$. In practice, we have $\sigma_{\varepsilon l}^2 \ll \sigma^2$ due to the fact that high SNR can be guaranteed in cooperative communications. On the other hand, linear manipulations in (8) will not change the SNR of the compressed signals $a_l \mathbf{\Phi}_l \mathbf{Q}_l \mathbf{s} + \mathbf{\Phi}_l \mathbf{w}_l$, and we can have

$$\mathbf{y}_l \approx \mathbf{\Phi}_l \mathbf{z}_l = a_l \mathbf{\Phi}_l \mathbf{Q}_l \mathbf{s} + \mathbf{\Phi}_l \mathbf{w}_l. \tag{9}$$

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III. DPD USING CS MEASUREMENTS

A. DPD Using Compressed Measurements

Generally, the measurements in (9) cannot be used directly for emitter positioning as the time-shift relationship between two receivers will be broken by the compressing matrix $\mathbf{\Phi}_l$. However, Salari et al. [18] show that it is possible to retain the time-shift relationship if a Hadamard compressing matrix is used.

A Hadamard matrix of order $N = 2^n (n = 1, 2, 3 \ldots)$, denoted by $\mathbf{H}_N$, is an $N \times N$ matrix whose elements are either $+1$ or -1 and its rows are mutually orthogonal [18]. By choosing M rows of the classical $\mathbf{H}_N$, an $M \times N$ Hadamard matrix can be obtained, denoted by $\mathbf{H}_{M\times N}$. It should be noted that the Hadamard matrix can only be used for compressing signals with dimension $N = 2^n$, $M = 2^m (n, m = 1, 2, 3, \ldots, m < n)$. On the other hand, we have the property

$$\begin{aligned}\mathbf{H}_{M\times N}^T \mathbf{H}_{M\times N} &= M \begin{bmatrix} \mathbf{I}_M & \mathbf{I}_M & \cdots & \mathbf{I}_M \\ \mathbf{I}_M & \mathbf{I}_M & \cdots & \mathbf{I}_M \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{I}_M & \mathbf{I}_M & \cdots & \mathbf{I}_M \end{bmatrix}_{N\times N} \\ &= M \begin{bmatrix} \mathbf{II}_{M\times N} \\ \mathbf{II}_{M\times N} \\ \vdots \\ \mathbf{II}_{M\times N} \end{bmatrix}_{N\times N} \end{aligned} \tag{10}$$

where $\mathbf{I}_M$ is an identity matrix with dimension $M \times M$ and $\mathbf{II}_{M\times N} = [\mathbf{I}_M \ \mathbf{I}_M \ \cdots \ \mathbf{I}_M]_{M\times N}$.

By (9) and (10), we have

$$\begin{aligned}&\mathbf{\Phi}_l^T \mathbf{y}_l = a_l \mathbf{\Phi}_l^T \mathbf{\Phi}_l \mathbf{Q}_l \mathbf{s} + \mathbf{\Phi}_l^T \mathbf{\Phi}_l \mathbf{w}_l \\ &= \mathbf{1}_{N/M} \otimes \left[M a_l \left(\mathbf{II}_{M\times N} \mathbf{Q}_l \mathbf{s} \right) + M \left(\mathbf{II}_{M\times N} \mathbf{w}_l \right) \right] \end{aligned} \tag{11}$$

where $\mathbf{1}_{N/M} = [1, 1, \ldots, 1]_{1\times(N/M)}^T$ and $\otimes$ denotes the Kronecker product. In detail, we have

$$\begin{aligned}&\mathbf{II}_{M\times N} \mathbf{Q}_l \mathbf{s} \\ &= \left[\sum_{i=0}^{N/M-1} s\left(iM\Delta + t_0 - \tau_l\right), \ldots, \sum_{i=0}^{N/M-1} \right. \\ &\quad \left. \times s\left(\left(iM + M - 1\right)\Delta + t_0 - \tau_l\right) \right]^T . \end{aligned} \tag{12}$$

On the other hand, we have

$$\begin{aligned}&\mathbf{II}_{M\times N} \mathbf{s} \\ &= \left[\sum_{i=0}^{N/M-1} s\left(iM\Delta + t_0\right), \ldots, \sum_{i=0}^{N/M-1} \right. \\ &\quad \left. \times s\left(\left(iM + M - 1\right)\Delta + t_0 \right) \right]. \end{aligned} \tag{13}$$

Equations (12) and (13) show that the time-shift relationship between $\mathbf{Q}_l \mathbf{s}$ and $\mathbf{s}$ is retained by the pair $\mathbf{II}_{M\times N} \mathbf{Q}_l \mathbf{s}$ and $\mathbf{II}_{M\times N} \mathbf{s}$. And clearly, the time-shift relationship between two receivers is retained after the received signals are compressed by a Hadamard matrix.

Similarly, in (6) and (7), we can express $\mathbf{II}_{M\times N} \mathbf{Q}_l \mathbf{s}$ as $\mathbf{Q}'_l \mathbf{s}'$, where

$$\mathbf{Q}'_l = \left(\mathbf{F}'\right)^H \mathbf{D}'_{\tau_l} \mathbf{F}' \tag{14a}$$

$$\tilde{\mathbf{m}} = \left[-\frac{M}{2}, \ldots, -1, 0, \ldots, \frac{M}{2} - 1 \right]^T \tag{14b}$$

$$\mathbf{F}' = \frac{1}{\sqrt{M}} \exp\left(-j \frac{2\pi}{M} \tilde{\mathbf{m}} \tilde{\mathbf{m}}^T \right) \tag{14c}$$

$$\mathbf{D}'_{\tau_l} = \operatorname{diag}\left\{ \exp\left(-j \frac{2\pi}{M} \tilde{\mathbf{m}} \frac{\tau_l}{\Delta} \right) \right\} \tag{14d}$$

$$\begin{aligned}\mathbf{s}' = \left[\sum_{i=0}^{N/M-1} s\left[iM\right], \sum_{i=0}^{N/M-1} s\left[iM + 1\right], \ldots, \sum_{i=0}^{N/M-1} \right. \\ \left. \times s\left[iM + M - 1\right] \right]^T . \end{aligned} \tag{15}$$

Let $\mathbf{II}_{M\times N} \mathbf{w}_l = \mathbf{w}'_l$, then we have

$$\mathbf{\Phi}_l^T \mathbf{y}_l = \mathbf{1}_{N/M} \otimes \left[M \left(a_l \mathbf{Q}'_l \mathbf{s}' + \mathbf{w}'_l \right) \right]. \tag{16}$$

For expression simplicity, we denote $a_\Sigma = \sqrt{\sum_{l=1}^L |a_l|^2}$. Equation (16) is redundant and duplicating $[M(a_l \mathbf{Q}'_l \mathbf{s}' + \mathbf{w}'_l)]$ by (N/M) times does not add new information, so we extract the first M elements of $\mathbf{\Phi}_l^T \mathbf{y}_l$ by

$$\mathbf{y}_{cl} = M \left(a_l \mathbf{Q}'_l \mathbf{s}' + \mathbf{w}'_l \right) = a_l M \mathbf{Q}'_l \mathbf{s}' + \boldsymbol{\varsigma}_l \tag{17}$$

where $\boldsymbol{\varsigma}_l$ is an independent circular complex Gaussian random vector whose probability density function (PDF) is $\boldsymbol{\varsigma}_l \sim \mathcal{N}(\mathbf{0}_M, \mathbf{\Lambda}_l)$, where $\mathbf{0}_M = [0, 0, \ldots, 0]_{1\times M}^T$ and $\mathbf{\Lambda}_l = NM\sigma^2 \mathbf{I}_M = \sigma_{cs}^2 \mathbf{I}_M$. By (17), we have

$$\mathbf{y}_c = M \mathbf{G} \mathbf{s}' + \boldsymbol{\varsigma} \tag{18}$$

where

$$\mathbf{y}_c = \left[\mathbf{y}_{c1}^T, \ldots, \mathbf{y}_{cL}^T \right]^T \tag{19a}$$

$$\mathbf{G} = \left[\left(a_1 \mathbf{Q}'_1\right)^T, \ldots, \left(a_L \mathbf{Q}'_L\right)^T \right]^T = \mathbf{A}\mathbf{Q} \tag{19b}$$

$$\mathbf{A} = \left[\operatorname{diag}\left(a_1, \ldots, a_L\right) \right] \otimes \mathbf{I}_M \tag{19c}$$

$$\mathbf{Q} = \left[\left(\mathbf{Q}'_1\right)^T \cdots \left(\mathbf{Q}'_L\right)^T \right]^T \tag{19d}$$

$$\boldsymbol{\varsigma} = \left[\left(\boldsymbol{\varsigma}_1\right)^T, \ldots, \left(\boldsymbol{\varsigma}_L\right)^T \right]^T. \tag{19e}$$

Define the unknown parameter $\boldsymbol{\theta} = \left[\mathbf{u}^T, \mathbf{a}^T, (\mathbf{s}')^T \right]^T$, where $\mathbf{a} = [a_1, \ldots, a_L]^T$, the logarithmic PDF of $\mathbf{y}_c$ is

$$\ln P\left(\mathbf{y}_c \,|\boldsymbol{\theta}\right) = c_0 - \frac{1}{2\sigma_{cs}^2} \left\| \mathbf{y}_c - M\mathbf{G}\mathbf{s}' \right\|^2. \tag{20}$$

Consequently, the maximum likelihood (ML) estimate of $\mathbf{s}'$ is

$$\mathbf{s}' = \frac{\mathbf{G}^H \mathbf{y}_c}{M a_\Sigma^2}. \tag{21}$$

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Substituting (21) into (20) and ignoring the terms irrelevant to $\mathbf{u}$ and $\mathbf{a}$, the objective function is formulated as

$$\begin{aligned} C(\mathbf{u},\mathbf{a}) &= a_{\Sigma}^{-2}\mathbf{y}_c^H\mathbf{G}\cdot\mathbf{G}^H\mathbf{y}_c \\ &= a_{\Sigma}^{-2}\sum_{i=1}^{L}\sum_{j=1}^{L}a_i a_j^*\left(\mathbf{y}_{ci}^H\mathbf{Q}'_i\mathbf{Q}'^H_j\mathbf{y}_{cj}\right) \\ &= \frac{\mathbf{a}^H}{\sqrt{\mathbf{a}^H\mathbf{a}}}R(\mathbf{u})\frac{\mathbf{a}}{\sqrt{\mathbf{a}^H\mathbf{a}}} \end{aligned} \tag{22}$$

where $R(\mathbf{u})$ is the cross-correlation matrix of size $L\times L$, whose (i,j)th entry is $(\mathbf{y}_{ci}^H\mathbf{Q}'_i\mathbf{Q}'^H_j\mathbf{y}_{cj})$.

For the case where the path attenuation $\mathbf{a}$ is unknown, it should be replaced with its ML estimate in (22), which is the scaled eigenvector corresponding to the largest eigenvalue of $R(\mathbf{u})$ [30]. Correspondingly, maximizing (22) is equivalent to maximizing [22]

$$C'(\mathbf{u}) = \lambda_{\max}(R(\mathbf{u})) \tag{23}$$

where $\lambda_{\max}(\cdot)$ represents the maximum eigenvalue of $(\cdot)$. Afterward, applying grid search over the region of interest or other optimization methods could obtain the estimate of $\mathbf{u}$ that maximizes (23).

B. RSS-Augmented DPD Using Compressed Measurements

In (23), the path attenuation factor a_l is treated as an unknown constant irrelevant to the source position. In the case of free-space propagation, which is common in practice, we could further have

$$a_l = \left(\frac{p_T G_T G_R \lambda^2}{(4\pi d_l)^2 L_l}\right)^{1/2} \tag{24}$$

where p_T is the transmitting power, G_T is the gain of the transmitting antenna, G_R is the gain of receiving antenna, λ is the wavelength, and d_l is the propagation distance. The term L_l is the system loss, which is irrelevant to the propagation distance but can be calibrated such that $L_1=\cdots=L_L=L_{\text{sys}}$.

For omnitransmitting and receiving antenna, we have

$$a_l = \frac{\kappa}{d_l} \tag{25}$$

where $\kappa=\sqrt{p_T\lambda^2/(4\pi)^2L_{\text{sys}}}$ is a constant. Substituting (25) into (22) and omitting the constant κ gives the RSS-augmented objective function

$$C_{\text{RSS}}(\mathbf{u}) = \frac{\mathbf{a}^H}{a_\Sigma}R(\mathbf{u})\frac{\mathbf{a}}{a_\Sigma} = \frac{\mathbf{d}^H}{\|\mathbf{d}\|}R(\mathbf{u})\frac{\mathbf{d}}{\|\mathbf{d}\|} \tag{26}$$

where $\mathbf{d}=[1/d_1,\ldots,1/d_L]^T$ and is clearly related to $\mathbf{u}$, where $d_l=c\tau_l$. Afterward, maximizing (26) gives the RSS-augmented estimate of $\mathbf{u}$. Usually, a grid search method is enough to maximize (26) when the dimension of $\mathbf{u}$ is low. In addition, particle filtering-based methods [26], [27] can be considered to accelerate the maximization process.

IV. PERFORMANCE ANALYSIS

This section first derives the emitter position CRLB for the scenario where RSS is simply ignored, then derives the RSS-augmented emitter position CRLB for the scenario of free-space propagation with omnitransmitting antenna.

A. CRLBS for DPD Using Compressed Measurements

In this case, the source signal waveform is unknown and $\mathbf{s}'$ can be expressed in terms of its real and imaginary parts as $\left[\text{Re}\{(\mathbf{s}')^T\},\text{Im}\{(\mathbf{s}')^T\}\right]^T$. And similarly, $\mathbf{a}$ can be expressed as $\left[\text{Re}\{\mathbf{a}^T\},\text{Im}\{\mathbf{a}^T\}\right]^T$. Define the composite unknown vector

$$\boldsymbol{\delta}_1 = \left[\text{Re}\left\{\left(\mathbf{s}'\right)^T\right\},\text{Im}\left\{\left(\mathbf{s}'\right)^T\right\},\text{Re}\left\{\mathbf{a}^T\right\},\text{Im}\left\{\mathbf{a}^T\right\},\mathbf{u}^T\right]^T. \tag{27}$$

From (18), the measurement vector $\mathbf{y}_c$ is a complex Gaussian random vector with mean $M\mathbf{G}\mathbf{s}'$ and covariance $\boldsymbol{\Lambda}=\sigma_{cs}^2\mathbf{I}_{LM}$. Consequently, the term $\mathbf{y}_c$ has a multivariate complex Gaussian PDF given by

$$\begin{aligned} p(\mathbf{y}_c;\boldsymbol{\delta}_1) = \frac{1}{\det(\pi\boldsymbol{\Lambda})}\exp\Big\{ &-\left(\mathbf{y}_c - M\mathbf{G}\mathbf{s}'\right)^H\boldsymbol{\Lambda}^{-1} \\ &\times\left(\mathbf{y}_c - M\mathbf{G}\mathbf{s}'\right)\Big\}. \end{aligned} \tag{28}$$

The Fisher information matrix (FIM) is then given by [26]

$$\begin{aligned} \mathbf{J}_{\delta_1} &= 2\text{Re}\left\{\left(\frac{\partial M\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_1}\right)^H\boldsymbol{\Lambda}^{-1}\left(\frac{\partial M\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_1}\right)\right\} \\ &= \frac{2M^2}{\sigma_{cs}^2}\text{Re}\left\{\left(\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_1}\right)^H\left(\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_1}\right)\right\} \end{aligned} \tag{29}$$

where

$$\begin{aligned} \frac{\partial\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_1} &= \left[\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\text{Re}\{\mathbf{s}'\}},\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\text{Im}\{\mathbf{s}'\}},\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\text{Re}\{\mathbf{a}\}},\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\text{Im}\{\mathbf{a}\}},\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\mathbf{u}}\right] \\ &= \left[\mathbf{G},j\mathbf{G},\mathbf{C},j\mathbf{C},\mathbf{D}\right] \end{aligned} \tag{30}$$

$$\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\text{Re}\{\mathbf{a}\}} = \mathbf{C} = \begin{pmatrix}\mathbf{Q}'_1\mathbf{s}' & \cdots & \mathbf{0}\\ \vdots & \cdots & \vdots\\ \mathbf{0} & \cdots & \mathbf{Q}'_L\mathbf{s}'\end{pmatrix} \tag{31}$$

$$\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\mathbf{u}} = \mathbf{D} = \mathbf{A}\frac{\partial(\mathbf{Q})}{\partial\mathbf{u}}\mathbf{s}' \tag{32}$$

where the detailed derivation of $\mathbf{D}$ can be referred to Appendix A. Consequently, the term $\mathbf{J}_{\delta_1}$ can be rewritten as

$$\begin{aligned} \mathbf{J}_{\delta_1} &= \frac{2M^2}{\sigma_{cs}^2}\text{Re}\left\{\left(\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_1}\right)^H\left(\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_1}\right)\right\} \\ &= \frac{2M^2}{\sigma_{cs}^2}\begin{pmatrix}\mathbf{V}_{11}^1 & \mathbf{V}_{12}^1\\ \mathbf{V}_{21}^1 & \mathbf{V}_{22}^1\end{pmatrix} \end{aligned} \tag{33}$$

where (34)–(36) shown at the bottom of the next page.

Using the partitioned matrix formula [31], the FIM of $\mathbf{u}$ is

$$\mathbf{J}_{\delta_1} = \frac{2M^2}{\sigma_{cs}^2}\left(\mathbf{V}_{22}^1 - \mathbf{V}_{21}^1\left(\mathbf{V}_{11}^1\right)^{-1}\mathbf{V}_{12}^1\right). \tag{37}$$

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Then, the CRLB of $\mathbf{u}$ in this case is

$$\mathrm{CRLB}_{\mathbf{u}}^{1}=\frac{\sigma_{cs}^{2}}{2M^{2}}\left(\mathbf{V}_{22}^{1}-\mathbf{V}_{21}^{1}\left(\mathbf{V}_{11}^{1}\right)^{-1}\mathbf{V}_{12}^{1}\right)^{-1}. \tag{38}$$

And the CRLB of localization error is

$$\mathrm{CRLB}_{\mathrm{DPD-CS}}=\sqrt{\mathrm{trace}\left(\mathrm{CRLB}_{\mathbf{u}}^{1}\right)}. \tag{39}$$

B. CRLBS for RSS-Augmented DPD Using Compressed Measurements

In this case, the composite unknown vector becomes

$$\boldsymbol{\delta}_{2}=\left[\mathrm{Re}\left\{\mathbf{s}^{T}\right\},\mathrm{Im}\left\{\mathbf{s}^{T}\right\},\mathbf{u}^{T}\right]^{T}. \tag{40}$$

The FIM is then given by

$$\mathbf{J}_{\boldsymbol{\delta}_{2}}=\frac{2M^{2}}{\sigma_{cs}^{2}}\mathrm{Re}\left\{\left(\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_{2}}\right)^{H}\left(\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_{2}}\right)\right\} \tag{41}$$

$$\begin{aligned}\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_{2}}&=\left[\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\mathrm{Re}\left\{\mathbf{s}'\right\}},\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\mathrm{Im}\left\{\mathbf{s}'\right\}},\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\mathbf{u}}\right]\\&=\left[\mathbf{G},\,j\mathbf{G},\,\mathbf{D}_{2}\right]\end{aligned} \tag{42}$$

$$\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\mathbf{u}}=\mathbf{D}_{2}=\frac{\partial\left(\mathbf{A}\mathbf{Q}\mathbf{s}'\right)}{\partial\mathbf{u}}=\frac{\partial\mathbf{A}}{\partial\mathbf{u}}\left(\mathbf{Q}\mathbf{s}'\right)+\mathbf{D} \tag{43}$$

where the derivation of $\mathbf{D}_2$ can be referred to Appendix B. Then, the term $\mathbf{J}_{\boldsymbol{\delta}_2}$ can be rewritten as

$$\begin{aligned}\mathbf{J}_{\boldsymbol{\delta}_{2}}&=\frac{2M^{2}}{\sigma_{cs}^{2}}\mathrm{Re}\left\{\left(\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_{2}}\right)^{H}\left(\frac{\partial\mathbf{G}\mathbf{s}'}{\partial\boldsymbol{\delta}_{2}}\right)\right\}\\&=\frac{2M^{2}}{\sigma_{cs}^{2}}\begin{pmatrix}\mathbf{V}_{11}^{2}&\mathbf{V}_{12}^{2}\\\mathbf{V}_{21}^{2}&\mathbf{V}_{22}^{2}\end{pmatrix}\end{aligned} \tag{44}$$

where

$$\mathbf{V}_{11}^{2}=\begin{bmatrix}\|\mathbf{a}\|\cdot\mathbf{I}_{M}&\mathbf{0}\\\mathbf{0}&\|\mathbf{a}\|\cdot\mathbf{I}_{M}\end{bmatrix} \tag{45}$$

$$\mathbf{V}_{12}^{2}=\begin{bmatrix}\mathrm{Re}\left(\mathbf{G}^{H}\mathbf{D}_{2}\right)\\\mathrm{Im}\left(\mathbf{G}^{H}\mathbf{D}_{2}\right)\end{bmatrix}=\left(\mathbf{V}_{12}^{2}\right)^{T} \tag{46}$$

$$\mathbf{V}_{22}^{2}=\mathrm{Re}\left(\mathbf{D}_{2}^{H}\mathbf{D}_{2}\right). \tag{47}$$

Using the partitioned matrix formula, the FIM of $\mathbf{u}$ is

$$\mathbf{J}_{\boldsymbol{\delta}_{2}}=\frac{2}{\sigma_{w}^{2}}\left(\mathbf{V}_{22}^{2}-\mathbf{V}_{21}^{2}\left(\mathbf{V}_{11}^{2}\right)^{-1}\mathbf{V}_{12}^{2}\right). \tag{48}$$

Then, the CRLB of $\mathbf{u}$ in this case is

$$\mathrm{CRLB}_{\mathbf{u}}^{2}=\frac{\sigma_{w}^{2}}{2M^{2}}\left(\mathbf{V}_{22}^{2}-\mathbf{V}_{21}^{2}\left(\mathbf{V}_{11}^{2}\right)^{-1}\mathbf{V}_{12}^{2}\right)^{-1}. \tag{49}$$

And the CRLB of localization error is

$$\mathrm{CRLB}_{\mathrm{RSS-DPD-CS}}=\sqrt{\mathrm{trace}\left(\mathrm{CRLB}_{\mathbf{u}}^{2}\right)}. \tag{50}$$

V. NUMERICAL INVESTIGATIONS

This section conducts several numerical experiments to evaluate the performance of the proposed DPD methods. We consider a scenario where one stationary source and three fixed receivers are present and the signals propagate in free space. The source is located at $\mathbf{u}=[14.5,\,102.6,\,10]^{T}$ (km) with the known altitude. The positions of the receivers are $\mathbf{p}_1=[40,\,1,\,10]^{T}$ (km), $\mathbf{p}_2=[0,\,0,\,10]^{T}$ (km), and $\mathbf{p}_3=[0,\,80,\,10]^{T}$ (km), respectively. The source emits signals with BPSK modulation and the bandwidth is 25 kHz. The carrier frequency is set to be $f_c=403$ MHz and the sampling frequency is $f_s=100$ kHz. A total of 1024 sampling points are collected per source location. For simplicity and without the loss of generality, we assume $\kappa=1$.

To gain more insights into the proposed DPD methods, three benchmarks are considered. The first benchmark is the classical DPD method without CS. The second benchmark is the two-step localization method based on the compressed measurements as considered in [18]. And the third benchmark is the two-step localization method based on the original uncompressed measurements. Fig. 1 presents the CRLBs of the proposed DPD methods as well as those of the benchmarks when the compressing rate is $M/N=1/2$ for different SNR conditions, where $\mathrm{CRLB}_{\mathrm{Two\text{-}Step}}$ refers to the CRLB of the third benchmark, $\mathrm{CRLB}_{\mathrm{Two\text{-}Step\text{-}CS}}$ refers to the CRLB of the second benchmark, and $\mathrm{CRLB}_{\mathrm{DPD}}$ refers to the CRLB of the first benchmark. In addition, based on 100 samples for each condition, Fig. 2 compares the statistical localization performance of the methods considered in Fig. 1.

$$\mathbf{V}_{11}^{1}=\begin{bmatrix}\|\mathbf{a}\|\cdot\mathbf{I}_{M}&\mathbf{0}&\mathrm{Re}\left(\mathbf{a}^{H}\otimes\mathbf{s}'\right)&-\mathrm{Im}\left(\mathbf{a}^{H}\otimes\mathbf{s}'\right)\\\mathbf{0}&\|\mathbf{a}\|\cdot\mathbf{I}_{M}&\mathrm{Im}\left(\mathbf{a}^{H}\otimes\mathbf{s}'\right)&\mathrm{Re}\left(\mathbf{a}^{H}\otimes\mathbf{s}'\right)\\\mathrm{Re}\left(\mathbf{a}\otimes\left(\mathbf{s}'\right)^{H}\right)&-\mathrm{Im}\left(\mathbf{a}\otimes\left(\mathbf{s}'\right)^{H}\right)&\mathbf{I}_{L}&\mathbf{0}\\\mathrm{Im}\left(\mathbf{a}\otimes\left(\mathbf{s}'\right)^{H}\right)&\mathrm{Re}\left(\mathbf{a}\otimes\left(\mathbf{s}'\right)^{H}\right)&\mathbf{0}&\mathbf{I}_{L}\end{bmatrix} \tag{34}$$

$$\mathbf{V}_{12}^{1}=\begin{bmatrix}\mathrm{Re}\left(\mathbf{G}^{H}\mathbf{D}\right)\\\mathrm{Im}\left(\mathbf{G}^{H}\mathbf{D}\right)\\\mathrm{Re}\left(\mathbf{C}^{H}\mathbf{D}\right)\\\mathrm{Im}\left(\mathbf{C}^{H}\mathbf{D}\right)\end{bmatrix}=\left(\mathbf{V}_{12}^{1}\right)^{T} \tag{35}$$

$$\mathbf{V}_{22}^{1}=\mathrm{Re}\left(\mathbf{D}^{H}\mathbf{D}\right). \tag{36}$$

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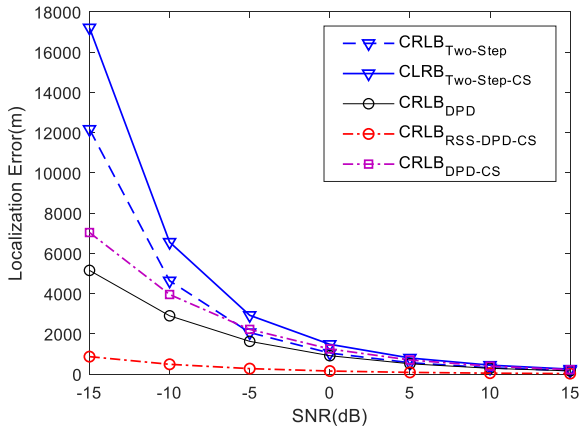


Fig. 1. CRLBs of the proposed DPD methods and three benchmarks.

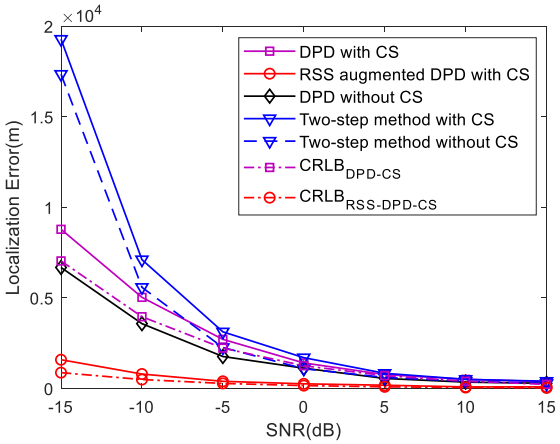


Fig. 2. Performance of the proposed DPD methods and three benchmarks.

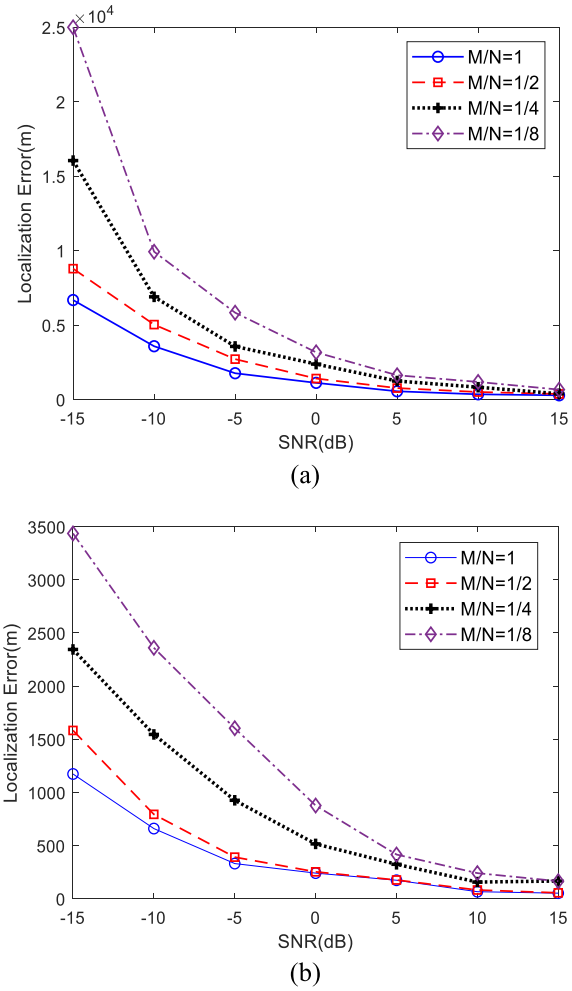


Fig. 3. Performance of the (a) proposed CS-based DPD method and (b) RSS-augmented CS-based DPD method for different compressing rates.

The results in Figs. 1 and 2 show that the proposed CS-based DPD method outperforms its two-step counterpart (the second benchmark), and the performance margin is significant when SNR is low. Interestingly, the CS-based DPD is even superior to the two-step method with uncompressed measurements (the third benchmark) when SNR is low and is only slightly inferior to it when SNR is larger than -5 dB. When the emitted signals are subject to free-space propagation, the RSS-augmented CS-based DPD method performs much better than the original version by using the additional position-related information in RSS. In addition, the performance of the proposed DPD methods slightly deviates from their CRLBs when SNR is low and gradually matches well with the CRLBs when SNR becomes high.

Fig. 3 summarizes the performance of the proposed methods for different compressing rates M/N and SNRs, where 100 samples are simulated for each case. It can be seen that the proposed DPD methods perform well at high SNRs even with low M/N values. In addition, it appears that the proposed DPD methods can perform well even at low SNRs by choosing a reasonably high M/N value (e.g., 1/2). It is noted that if we choose $M/N = 1/2$, the performance of the proposed DPD methods is within about 3 dB of the no CS case over all SNRs while reducing the communication amount to a rate of M/N. Therefore, the proposed DPD methods are appropriate for reducing the communication burden while maintaining reasonable localization accuracy.

VI. CONCLUSION

We have developed a CS-based DPD method that eliminates both the complex reconstruction step and the TDOA parameter extraction step. The method can estimate the source location directly in the compressive measurement domain. In addition, an RSS-augmented DPD method is proposed to further improve the localization accuracy when signals are subject to free-space propagation. The CRLBs for the emitter position of both DPD methods are derived for performance evaluation. The proposed CS-based DPD method performs comparably with the classical DPD method without CS at high SNRs. With a reasonable compressing rate, the CS-based DPD method can also perform well when SNR is low. When the emitted signals are subject to free-space propagation, better performance can be achieved by the RSS-augmented CS-based DPD method. It would be interesting to study the feasibility of extending the proposed method to source localization with

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moving receivers. In addition, when the signals are subject to both free-space propagation and fading, it is necessary to generalize the localization objective function.

APPENDIX A
DERIVATION OF $\mathbf{D}$

From the definition of $\mathbf{A}$ and $\mathbf{Q}$, we have

$$\mathbf{D} = \mathbf{A}\frac{\partial(\mathbf{Q})}{\partial \mathbf{u}}\mathbf{s}' = \left[\left(a_1\frac{\partial\left(\mathbf{Q}'_1\mathbf{s}'\right)}{\partial \mathbf{u}}\right)^H, \ldots, \times\left(a_L\frac{\partial\left(\mathbf{Q}'_L\mathbf{s}'\right)}{\partial \mathbf{u}}\right)^H\right]^H. \tag{51}$$

From the definition of $\mathbf{Q}'_l$ in (14a), we have

$$\frac{\partial\left(\mathbf{Q}'_l\mathbf{s}'\right)}{\partial \mathbf{u}} = \frac{\partial\left(\mathbf{Q}'_l\mathbf{s}'\right)}{\partial \tau_l}\frac{\partial \tau_l}{\partial \mathbf{u}} \tag{52}$$

where

$$\frac{\partial\left(\mathbf{Q}'_l\mathbf{s}'\right)}{\partial \tau_l} = -j\frac{2\pi}{M\Delta}\left(\mathbf{F}'\right)^H\mathbf{D}'_{\tau_l}\mathrm{diag}\left\{\mathbf{m}\right\}\mathbf{F}'\mathbf{s}' \tag{53}$$

$$\frac{\partial \tau_l}{\partial \mathbf{u}} = \frac{(\mathbf{u}-\mathbf{p}_l)^T}{c\left\|\mathbf{u}-\mathbf{p}_l\right\|}. \tag{54}$$

Combining (51)–(54) gives the detailed formulation of $\mathbf{D}$.

APPENDIX B
DERIVATION OF $\mathbf{D}_2$

From the definition of $\mathbf{D}_2$ in (43), we have

$$\mathbf{D}_2 = \frac{\partial \mathbf{A}}{\partial \mathbf{u}}\left(\mathbf{Q}\mathbf{s}'\right) + \mathbf{D} \tag{55}$$

where

$$\frac{\partial \mathbf{A}}{\partial \mathbf{u}} = \left[\mathrm{diag}\left(\frac{\partial a_1}{\partial \mathbf{u}}, \ldots, \frac{\partial a_L}{\partial \mathbf{u}}\right)\right] \otimes \mathbf{I}_M. \tag{56}$$

From the definition of a_l in (25), we further have

$$\frac{\partial a_l}{\partial \mathbf{u}} = -\kappa\frac{(\mathbf{u}-\mathbf{p}_l)^T}{\left\|\mathbf{u}-\mathbf{p}_l\right\|^3}. \tag{57}$$

Combining (55)–(57) gives the detailed formulation of $\mathbf{D}_2$.

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